

$$C_{\phi}^{(12)} [{}^3_1 i_1 i_2] \rightarrow \frac{1}{16\pi^2} \left(\frac{1}{3} m_2 \tilde{\mu} g_2^4 LF_{2,2,\emptyset}[m_2, \tilde{\mu}] \delta_{i_1 i_2} - \frac{1}{6} m_2 \tilde{\mu} g_2^4 LF_{3,2,-1}[m_2, \tilde{\mu}] \delta_{i_1 i_2} + \right. \\ \frac{1}{8} \tilde{\mu} g_2^2 \overline{y_e^{pr}} a_e^{pr} LF_{4,1,-1}[m_l^P, m_e^r] \delta_{i_1 i_2} - \frac{1}{6} \tilde{\mu} g_2^2 \overline{y_e^{pr}} a_e^{pr} LF_{5,1,-2}[m_l^P, m_e^r] \delta_{i_1 i_2} + \\ \frac{3}{8} \tilde{\mu} g_2^2 \overline{y_d^{pr}} a_d^{pr} LF_{4,1,-1}[m_q^P, m_d^r] \delta_{i_1 i_2} - \frac{1}{2} \tilde{\mu} g_2^2 \overline{y_d^{pr}} a_d^{pr} LF_{5,1,-2}[m_q^P, m_d^r] \delta_{i_1 i_2} + \\ \frac{1}{4} \tilde{\mu} g_2^2 \overline{y_u^{pr}} a_u^{pr} LF_{2,2,\emptyset}[m_q^P, m_u^r] \delta_{i_1 i_2} - \frac{1}{8} \tilde{\mu} g_2^2 \overline{y_u^{pr}} a_u^{pr} LF_{3,2,-1}[m_q^P, m_u^r] \delta_{i_1 i_2} - \\ \frac{1}{4} \tilde{\mu} g_2^2 \overline{y_u^{pr}} a_u^{pr} LF_{3,1,\emptyset}[m_u^r, m_q^P] \delta_{i_1 i_2} - \frac{1}{4} \tilde{\mu} g_2^2 \overline{y_u^{pr}} a_u^{pr} LF_{3,2,-1}[m_u^r, m_q^P] \delta_{i_1 i_2} + \\ \frac{3}{4} \tilde{\mu} g_2^2 \overline{y_u^{pr}} a_u^{pr} LF_{4,1,-1}[m_u^r, m_q^P] \delta_{i_1 i_2} - \frac{1}{2} \tilde{\mu} g_2^2 \overline{y_u^{pr}} a_u^{pr} LF_{5,1,-2}[m_u^r, m_q^P] \delta_{i_1 i_2} + \\ \frac{1}{8} m_1 \tilde{\mu} g_1^2 g_2^2 LF_{4,1,-1}[\tilde{\mu}, m_1] \delta_{i_1 i_2} - \frac{1}{6} m_1 \tilde{\mu} g_1^2 g_2^2 LF_{5,1,-2}[\tilde{\mu}, m_1] \delta_{i_1 i_2} - \\ \frac{1}{3} m_2 \tilde{\mu} g_2^4 LF_{3,1,\emptyset}[\tilde{\mu}, m_2] \delta_{i_1 i_2} - \frac{1}{3} m_2 \tilde{\mu} g_2^4 LF_{3,2,-1}[\tilde{\mu}, m_2] \delta_{i_1 i_2} + \\ \frac{7}{8} m_2 \tilde{\mu} g_2^4 LF_{4,1,-1}[\tilde{\mu}, m_2] \delta_{i_1 i_2} - \frac{1}{2} m_2 \tilde{\mu} g_2^4 LF_{5,1,-2}[\tilde{\mu}, m_2] \delta_{i_1 i_2} - \\ \frac{1}{8} m_1 \tilde{\mu} g_1^2 \overline{y_e^{i2p}} y_e^{i1p} LF_{2,1,1,\emptyset}[m_1, m_e^P, m_l^{i2}] - \frac{1}{16} m_1 \tilde{\mu} g_1^2 \overline{y_e^{i2p}} y_e^{i1p} LF_{2,2,1,-1}[m_1, m_e^P, m_l^{i2}] + \\ \frac{1}{8} m_1 \tilde{\mu} g_1^2 \overline{y_e^{i2p}} y_e^{i1p} LF_{3,1,1,-1}[m_1, m_e^P, m_l^{i2}] + \frac{1}{4} m_1 \tilde{\mu} g_1^2 \overline{y_e^{i2p}} y_e^{i1p} LF_{2,1,1,\emptyset}[m_1, m_e^P, \tilde{\mu}] - \\ \frac{1}{4} m_1 \tilde{\mu} g_1^2 \overline{y_e^{i2p}} y_e^{i1p} LF_{3,1,1,-1}[m_1, m_e^P, \tilde{\mu}] + \frac{1}{16} m_1 \tilde{\mu} g_1^2 \overline{y_e^{i2p}} y_e^{i1p} LF_{2,2,1,-1}[m_1, m_l^{i2}, m_e^P] - \\ \frac{3}{16} m_1 \tilde{\mu} g_1^2 \overline{y_e^{i2p}} y_e^{i1p} LF_{2,2,1,-1}[m_1, \tilde{\mu}, m_e^P] + \frac{1}{16} m_1 \tilde{\mu} g_1^2 \overline{y_e^{i2p}} y_e^{i1p} LF_{2,2,1,-1}[m_1, \tilde{\mu}, m_l^{i2}] + \\ \frac{1}{16} m_2 \tilde{\mu} g_2^2 \overline{y_e^{i2p}} y_e^{i1p} LF_{2,2,1,-1}[m_2, \tilde{\mu}, m_e^P] - \frac{1}{4} m_2 \tilde{\mu} g_2^4 LF_{2,2,1,-1}[m_2, \tilde{\mu}, m_l^{i1}] \delta_{i_1 i_2} - \\ \frac{3}{16} m_2 \tilde{\mu} g_2^2 \overline{y_e^{i2p}} y_e^{i1p} LF_{2,2,1,-1}[m_2, \tilde{\mu}, m_l^{i2}] - \frac{1}{8} m_1 \tilde{\mu} g_1^2 \overline{y_e^{i2p}} y_e^{i1p} LF_{2,1,1,\emptyset}[\tilde{\mu}, m_1, m_l^{i2}] + \\ \frac{1}{8} m_1 \tilde{\mu} g_1^2 \overline{y_e^{i2p}} y_e^{i1p} LF_{3,1,1,-1}[\tilde{\mu}, m_1, m_l^{i2}] + \frac{3}{8} m_2 \tilde{\mu} g_2^2 \overline{y_e^{i2p}} y_e^{i1p} LF_{2,1,1,\emptyset}[\tilde{\mu}, m_2, m_l^{i2}] - \\ \frac{3}{8} m_2 \tilde{\mu} g_2^2 \overline{y_e^{i2p}} y_e^{i1p} LF_{3,1,1,-1}[\tilde{\mu}, m_2, m_l^{i2}] + \frac{1}{16} \tilde{\mu} g_1^2 \overline{y_e^{i2p}} a_e^{i1p} LF_{1,1,1,1,\emptyset}[m_1, m_e^P, m_l^{i1}, \tilde{\mu}] - \\ \frac{1}{32} \tilde{\mu} g_1^2 \overline{y_e^{i2p}} a_e^{i1p} LF_{2,1,1,1,-1}[m_1, m_e^P, m_l^{i1}, \tilde{\mu}] + \frac{1}{16} m_1 \tilde{\mu} g_1^2 \overline{y_e^{i2p}} y_e^{i1p} \\ LF_{1,1,1,1,\emptyset}[m_1, m_e^P, m_l^{i2}, \tilde{\mu}] + \frac{1}{32} m_1 \tilde{\mu} g_1^2 \overline{y_e^{i2p}} y_e^{i1p} LF_{2,1,1,1,-1}[m_1, m_e^P, m_l^{i2}, \tilde{\mu}] + \\ \frac{1}{16} \tilde{\mu} g_2^2 \overline{y_e^{i2p}} a_e^{i1p} LF_{1,1,1,1,\emptyset}[m_2, m_e^P, m_l^{i1}, \tilde{\mu}] - \frac{1}{32} \tilde{\mu} g_2^2 \overline{y_e^{i2p}} a_e^{i1p} \\ LF_{2,1,1,1,-1}[m_2, m_e^P, m_l^{i1}, \tilde{\mu}] + \frac{1}{16} m_2 \tilde{\mu} g_2^2 \overline{y_e^{i2p}} y_e^{i1p} LF_{1,1,1,1,\emptyset}[m_2, m_e^P, m_l^{i2}, \tilde{\mu}] + \\ \frac{1}{32} m_2 \tilde{\mu} g_2^2 \overline{y_e^{i2p}} y_e^{i1p} LF_{2,1,1,1,-1}[m_2, m_e^P, m_l^{i2}, \tilde{\mu}] - \frac{1}{16} \tilde{\mu} g_1^2 \overline{y_e^{i2p}} a_e^{i1p} \\ LF_{2,1,1,1,-1}[m_e^P, m_1, m_l^{i1}, m_l^{i2}] + \frac{1}{32} \tilde{\mu} g_1^2 \overline{y_e^{i2p}} a_e^{i1p} LF_{2,1,1,1,-1}[m_e^P, m_1, m_l^{i1}, \tilde{\mu}] - \\ \frac{1}{32} m_1 \tilde{\mu} g_1^2 \overline{y_e^{i2p}} y_e^{i1p} LF_{2,1,1,1,-1}[m_e^P, m_1, m_l^{i2}, \tilde{\mu}] + \frac{1}{16} \tilde{\mu} g_2^2 \overline{y_e^{i2p}} a_e^{i1p} \\ LF_{2,1,1,1,-1}[m_e^P, m_2, m_l^{i1}, m_l^{i2}] + \frac{1}{32} \tilde{\mu} g_2^2 \overline{y_e^{i2p}} a_e^{i1p} LF_{2,1,1,1,-1}[m_e^P, m_2, m_l^{i1}, \tilde{\mu}] - \\ \frac{1}{32} m_2 \tilde{\mu} g_2^2 \overline{y_e^{i2p}} y_e^{i1p} LF_{2,1,1,1,-1}[m_e^P, m_2, m_l^{i2}, \tilde{\mu}] - \frac{1}{32} \tilde{\mu} g_1^2 \overline{y_e^{i2p}} a_e^{i1p} \\ LF_{2,1,1,1,-1}[m_l^{i1}, m_1, m_e^P, \tilde{\mu}] - \frac{1}{32} \tilde{\mu} g_2^2 \overline{y_e^{i2p}} a_e^{i1p} LF_{2,1,1,1,-1}[m_l^{i1}, m_2, m_e^P, \tilde{\mu}] + \\ \frac{1}{32} m_1 \tilde{\mu} g_1^2 \overline{y_e^{i2p}} y_e^{i1p} LF_{2,1,1,1,-1}[m_l^{i2}, m_1, m_e^P, \tilde{\mu}] + \frac{1}{32} m_2 \tilde{\mu} g_2^2 \overline{y_e^{i2p}} y_e^{i1p} \\ LF_{2,1,1,1,-1}[m_l^{i2}, m_2, m_e^P, \tilde{\mu}] + \frac{1}{32} \tilde{\mu} g_1^2 \overline{y_e^{i2p}} a_e^{i1p} LF_{2,1,1,1,-1}[\tilde{\mu}, m_1, m_e^P, m_l^{i1}] - \\ \frac{1}{32} m_1 \tilde{\mu} g_1^2 \overline{y_e^{i2p}} y_e^{i1p} LF_{2,1,1,1,-1}[\tilde{\mu}, m_1, m_e^P, m_l^{i2}] + \frac{1}{32} \tilde{\mu} g_2^2 \overline{y_e^{i2p}} a_e^{i1p} \\ LF_{2,1,1,1,-1}[\tilde{\mu}, m_2, m_e^P, m_l^{i1}] - \frac{1}{32} m_2 \tilde{\mu} g_2^2 \overline{y_e^{i2p}} y_e^{i1p} LF_{2,1,1,1,-1}[\tilde{\mu}, m_2, m_e^P, m_l^{i2}])$$